In general, there are three main parts you want to consider with backends, whether that is concerning performance or costs (see section 5.6 for more):

- *Ingestion*—How the signals arrive at the backend. Various protocols and formats exist, such as OpenTelemetry's OTLP or the Prometheus formats.
- *Storage*—Usually, there's the expectation that the signals ingested are stored for a longer time period (weeks, months, or even years), allowing for historical query and data analytics. But this doesn't always have to be the case, as pure in-memory solutions exist (e.g., CNCF Pixie) and have their uses, such as live cluster debugging, keeping only hours of (volatile) data around.
- *Query*—The more exact and common term we use for *asking a backend a question* is to *query the backend*. What exactly *query* means can differ from backend to backend and can range from a declarative style, such as SQL, to an API call, such as is the case with many tracing backends. You will also often come across domain-specific languages (DSLs), such as PromQL, and you'll see several examples of these DSLs in use throughout this chapter. With this in mind, let's move on to the first, and probably most widely adopted, kind of backend, for all sorts of logs.

5.2 Backend destinations for logs

In this section, we will take a cursory look at backend destinations that you can use to ingest logs and query them. As we discussed in chapter 2, logs are timestamped, typically capturing events; have a (hopefully structured) payload; and are primarily meant for human consumption. The sheer number of different log types and formats makes it sometimes difficult to figure out what the optimal backend is. Usually, it depends heavily on the use case and your preferences, such as open source and open standards first, or you might prefer a managed offering over a DIY approach (or the other way round). Having said that, and of course this also applies to the rest of the backends for metrics and distributed traces, let's see what we have here.

5.2.1 Cloud providers

Every cloud provider has, at the very least, a service in their offerings that allows you to land logs—more specially, that is (focusing on the big three hyperscalers):

- Amazon CloudWatch Logs (http://mng.bz/yQjp) organizes different kinds of logs in log groups that, in turn, comprise one or more log streams (think of a never-ending series of events), which are made up of log events. It uses a proprietary way to ingest the logs and offers the embedded metric format (EMF; http://mng.bz/nWa5), a format that allows you to embed custom metrics alongside logs. In other words, EMF is more event-centric (supporting high cardinality) than a pure log format. You can use CloudWatch Logs Insights, which provides a purpose-built query language to query your logs.